AI-Powered Full Stack Developer Program

HOMEWORK SHEET

Week 2 · Day 2

Semantic HTML, Accessibility, Media → CSS Basics

Homework HW1–HW4: tasks, time boxes, submit checklist, self-check and interview prep

Momo House · TSI Canteen · Aarav · Sunita · Didi · Rajan

Stage 1, AI as Tutor: AI may explain; you type every line.

Tech Skills Institute (TSI) Nepal · Kathmandu

Week 2 · Day 2 Homework: Semantic HTML, Accessibility, Media → CSS Basics

File: homework/W2D2-Homework-Sheet.md

TSI Nepal · AI-Powered Full Stack Developer Program · Week 2, Day 2 · Layer: Frontend

Tonight in one sentence: you repair a haunted portfolio, then give your own page a semantic skeleton and a real outfit, and test yourself on why each rule works.

Core time: about 2.5 hours. **HW1 and HW3 are the must-do (1 hour).** Do HW2 if you can (about 1.5 hours). **Stretch:** HW4 adds about 30 minutes. It is optional.

Before you start (5 minutes)

1. Open VS Code. **File → Open Folder → `W2D2`.** Open the folder, not a single file.
2. Live Server and HTMLHint must be installed (`Ctrl+Shift+X`). Open the **Problems** panel (`Ctrl+Shift+M`).
3. In Chrome, open **F12**. You will live in two places tonight:
   - **Elements → Styles**: which CSS rules reached this element, and which are crossed out.
   - **Elements → Computed** (the box diagram): content, padding, border, margin.
4. Keep `notes/11-day2-cheatsheet.md` and `notes/09-debugging-css.md` open in a second tab.

CSS fails silently too. A wrong rule is not an error message. The browser just drops it. Your job is to notice what is missing. F12 → Styles shows dropped declarations with a yellow ⚠ or a line through them.

The plan for tonight

#	Id	Title	Type	Time	Folder
1	**D2-HW1**	Haunted portfolio repair	bug-hunt, 12 bugs	40 min (must-do)	`homework/D2-HW1-haunted-portfolio-repair/`
2	**D2-HW2**	Style your portfolio	build	about 1.5 h (if you can)	`homework/D2-HW2-style-your-portfolio/`
3	**D2-HW3**	Self-check & interview prep	questions	20 min (must-do)	`homework/D2-HW3-Self-Check-and-Interview-Prep.md`
4	**D2-HW4**	Patan Craft Shop replay (**stretch**)	bug-hunt, 10 bugs	+30 min	`homework/D2-HW4-patan-craft-shop-replay/`

Do them in this order. Short on time? Do HW1 and HW3 first; they are the must-do. HW1 trains your eye on HTML *and* CSS bugs together. HW2 is your own page, so it goes in your portfolio. HW3 checks you can explain it. HW4 is extra practice.

D2-HW1 · Haunted portfolio repair (40 min, must-do)

The story. Rajan built a 2-page portfolio (`index.html` and `projects.html`) with one shared stylesheet. Something haunts it: the page looks almost unstyled, the nav is a list, the cards overflow, and the footer ignores its colours. There are **12 bugs**.

Files: `index.html`, `projects.html`, `css/style.css`, `images/` (Rajan's avatar and two project pictures).

What you practise:

- linking a stylesheet (the path!)

- semantic rules: one `<main>`, sections need headings
- `alt` on images
- display values, class and id matching (case matters)
- units, colours, border shorthand
- the box model and `box-sizing`
- a missing `}`

How to work:

1. Read the folder's `README.md`. It has a **"fix this first"** note. Until the stylesheet loads, no CSS bug is visible.
2. Open both pages with Live Server.
3. Right-click anything that looks wrong → **Inspect**. In **Styles**, look for crossed-out or ⚠ lines. In **Computed**, read the box diagram.
4. Run Lighthouse → Accessibility on `index.html` (F12 → Lighthouse, works offline).

Done when:

- ☐ Both pages are styled: horizontal nav, cards side by side without overflow, coloured footer.
- ☐ 0 problems in the Problems panel for both HTML files.
- ☐ Lighthouse Accessibility is **90 or more** on `index.html`.
- ☐ You wrote down every bug: file, what was wrong, how you saw it (V, D(HH), D(LH), D(W3C) or R).

Scoreboard: `I found __ / 12. The bug that fooled me longest: _____`

D2-HW2 · Style your portfolio (about 1.5 h, if you can) 🔨

The story. This is *your* page, not Rajan's. Take your **Week 1 About Me page** (or your **D1-10 Kathmandu gallery**) and turn it into a small, styled portfolio. Copy it into a new folder `my-portfolio/` inside `homework/D2-HW2-style-your-portfolio/`, and add a second page (for example `about.html`), so you have **2 pages that share one stylesheet**: the "school uniform".

Open `homework/D2-HW2-style-your-portfolio/README.md` and `CHECKLIST.md`. The checklist has **12 required items**. In short:

1. `header`, `nav`, `main` and `footer` (semantic skeleton)
2. a horizontal nav (`display: inline-block` and `list-style: none`)
3. one box-model card (width, padding, border, margin, `box-sizing: border-box`)
4. font sizes in `rem`
5. a 3-colour hex palette
6. a `:hover` state
7. every `img` has a good `alt`
8. a `<label>` on every input
9. passes HTMLHint (0 problems)
10. Lighthouse Accessibility ≥ 90
11. works from the `W2D2` folder
12. every link resolves

Tips:

- Start with the HTML skeleton. Style only when the skeleton is right ("skeleton first, outfit second").
- Put the link in the `<head>` of every page: `<link rel="stylesheet" href="css/style.css">`.
- Change one thing, save, look. Live Server reloads for you.
- When something won't change, inspect it. Is your rule in Styles? Is it crossed out? Who won?

Done when: every one of the 12 boxes in `CHECKLIST.md` is ticked, and you can explain every CSS line.

Bonus: add a third page that uses the same stylesheet with no new CSS, or try D2-FF6 and compare your CSS with Bootstrap.

D2-HW3 · Self-check & interview prep (20 min, must-do)

Open `homework/D2-HW3-Self-Check-and-Interview-Prep.md`.

1. **Part 1, self-check (10 questions):** answer from memory first. Mark ✔ / ~ / ✗.
2. **Part 2, interview questions (14 questions, theory + small code):** say the theory answers **out loud** (20–40 seconds each). Write the code answers, then check them in the browser.
3. The answer key is shared by your instructor after you try. Re-read the notes for every ✗.

Done when: you have an answer for all 24 questions, and a list of your ✗ topics.

D2-HW4 · Patan Craft Shop replay (stretch, +30 min)

Optional. Do it only after HW1–HW3.

The story. A craft shop near Patan Durbar Square has a one-page site with a video and a stylesheet. It has **10 bugs**. They are the Day 2 bugs we didn't have time for in class.

Files: `index.html`, `css/style.css`, `media/momo-steam.mp4`, `images/` (craft shop, video poster).

What you practise: the head for search engines and phones (Lighthouse finds one; the device toolbar shows the other), video sources, id case, invalid values (too many, too few), widths that are too wide, heights that are too small, and hiding things the right way.

Hint: scroll sideways. Resize the window. Press play on both videos. Run Lighthouse. Try the phone view (F12 → Ctrl+Shift+M).

Done when: both videos play, nothing scrolls sideways, no text spills out of a box, Lighthouse finds no head problems, and the phone view is not tiny.

Scoreboard: `I found __ / 10. The new thing I learnt: _____`

Submit checklist (before the next class)

Zip each homework folder and send it the way your instructor asks. Also bring your `W2D2` folder on your laptop.

- ☐ **HW1:** Rajan's 2 pages are fixed; 0 HTMLHint problems; Lighthouse Accessibility ≥ 90 on `index.html`.
- ☐ **HW1:** my bug list is in `homework/D2-HW1-haunted-portfolio-repair/my-bugs.md`.
- ☐ **HW2:** my portfolio is in `homework/D2-HW2-style-your-portfolio/my-portfolio/` (2 HTML pages + `css/style.css` + `images/`).
- ☐ **HW2:** all 12 boxes in `CHECKLIST.md` are ticked, and I wrote my Lighthouse score next to item 10.
- ☐ **HW3:** I answered all 24 questions and wrote my ✗ list at the bottom of the HW3 file.
- ☐ *(Stretch)* **HW4:** the Patan Craft Shop page is fixed.
- ☐ I typed every line myself and can explain every line.
- ☐ **One volunteer** shows their HW2 portfolio at the start of the next class.

AI rule for this week

Stage 1, AI as Tutor. AI may **explain** a concept or an error. **You type every line.** Try first, then ask.

A good prompt for tonight:

> "My three cards are 300px wide inside a 960px row, but the third one wraps to a new line. I have padding 20px and a 5px border. Explain how the box model adds these up, and what `box-sizing` changes. Don't write my CSS."

A bad prompt: "Write CSS for my portfolio." (It would not be your portfolio, and you could not explain it in an interview.)

Stuck?

1. Re-read the folder's `README.md` hints.
2. Open the matching notes file: semantic tags → `notes/01`, alt/labels/Lighthouse → `02`, media → `03`, linking CSS → `04`, selectors → `05`, units/colours/fonts → `06`, box model → `07`, display → `08`, "my CSS is ignored" → `09`.
3. Inspect the element. Read Styles (crossed out?) and Computed (the box).
4. Ask a classmate to *point* (not type).

Remember the One Rule: you may only ship code you can explain, line by line. Coming next: CSS from basics to advanced (sessions 3 / 3.2): flexbox, grid, positioning and responsive design.

D2-HW1 · Haunted portfolio repair

File: homework/D2-HW1-haunted-portfolio-repair/README.md

Time: 40 min (at home, must-do) · **Type:** Bug-hunt

Goal: Rajan's portfolio is haunted: semantic mistakes and CSS mistakes together. Mend both pages and the stylesheet.

Open the folder in VS Code, then open `index.html` with Live Server.

Task

1. Open both pages: `index.html` and `projects.html`. Compare how they look.
2. Mend the home page's `<link>` first (see below). Then go through the rest.
3. Check each page with your eyes, HTMLHint (Problems panel), F12 → Elements and Lighthouse.
4. Write each bug you found in a short list: file, line, what was wrong, how you mended it.

The bugs

There are 12 bugs. Find at least 10.

> **Fix this first:** bug 1, the `<link>` in `index.html`. Until the home page loads its CSS, you can't see the CSS bugs there.

Hints, one per bug type:

- **The link:** where is the CSS file?
- **Semantic:** how many `<main>`? Does every `<section>` have a heading?
- **Images:** every meaningful image needs words.
- **Values:** units, hex digits, typos in `display`.
- **Selectors:** capital letters; `#` is for ids, `.` is for classes.
- **Box model:** do two cards fit side by side? Does every border have a style?
- **Braces:** every `{` needs a `}`.

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Done looks like

Both pages share one look: a navy bar with a horizontal white menu, a centred white hero with a round photo, two project cards side by side with grey borders, and a navy footer. HTMLHint shows no problems.

Scoreboard

```
I found __ / 12. The reveal taught me: ________
```

Stuck?

1. Do the easy ones first: anything you can see. Then the semantic ones (read the code).
2. F12 → Elements → Styles shows crossed-out declarations and missing rules.
3. Ask the AI tutor to explain an error message, not to mend the file for you.

Fast?

Start **D2-HW2** (style your own portfolio).

D2-HW2 · Style your portfolio

File: homework/D2-HW2-style-your-portfolio/README.md

Time: about 1.5 h (at home, if you can; HW1 and HW3 come first) · **Type:** 🔨 Build

Goal: Take a page you made yourself and make it semantic, accessible, and dressed in one shared stylesheet.

Start from

Pick **one**:

- your Week 1 **About Me** page, or
- your **D1-10 Kathmandu gallery** from Day 1.

Copy it into a new folder `my-portfolio/` inside `W2D2/homework/D2-HW2-style-your-portfolio/`. Make a second page (for example `about.html` or `projects.html`) so you have **2 pages that share one stylesheet**.

Task

1. **Semantic skeleton:** replace `div`s with `header`, `nav`, `main`, `section`/`article`, `footer`. One `<main>` per page.
2. **Accessible:** `lang="en"`, a real `<title>`, alt on every image, a label on every input, no heading jumps.
3. **One stylesheet:** create `css/style.css` and link it from **both** pages (the school uniform).
4. **Style it** with everything from today: a 3-colour hex palette, a font stack with a fallback, `rem` sizes, `line-height`, a horizontal nav, one box-model card, a `:hover` state.
5. Go through `CHECKLIST.md`: all 12 items.

Done looks like

Both pages look like one website. HTMLHint shows no problems. Lighthouse Accessibility is 90 or more. You can explain every line of your CSS (the One Rule).

Stuck?

1. Look at your D2-15 Momo House stylesheet: the same 5 checkpoints work for your portfolio.
2. Read `notes/11-day2-cheatsheet.md`.
3. Ask the AI tutor to *explain* a property. Type the code yourself.

Fast?

Add a third page, or try D2-FF6 and compare your hand-made CSS with Bootstrap.

D2-HW2 · Style your portfolio — CHECKLIST (12 items)

File: homework/D2-HW2-style-your-portfolio/CHECKLIST.md

Tick each box before you submit. The model answer (Rajan's portfolio) is in `homework/_solutions/D2-HW2-style-your-portfolio/` and will be shared after you submit.

#	Required	How to check	✔
1	`<header>`, `<nav>`, `<main>` and `<footer>` on every page (one `<main>` per page)	F12 → Elements	☐
2	A horizontal nav: no bullets, links side by side (`list-style: none` + `display: inline-block`)	your eyes	☐
3	One box-model card: width, padding, border, `border-radius`, and `box-sizing: border-box`	F12 → Computed box diagram	☐
4	Font sizes in `rem` (no `px` font sizes)	search `font-size` in your CSS	☐
5	A 3-colour palette written as hex codes, used across the site	read your CSS	☐
6	At least one `:hover` state (nav links or a button)	move the mouse	☐
7	Every `<img>` has an `alt` (decorative images: `alt=""`)	HTMLHint / Lighthouse	☐
8	Every `<input>` has a `<label>` (if you have a form)	click the label text: the cursor jumps into the box	☐
9	HTMLHint: zero problems on every page	Problems panel (Ctrl+Shift+M)	☐
10	Lighthouse Accessibility score **90 or more**	F12 → Lighthouse → Accessibility	☐
11	Works from the `W2D2` folder: relative paths only (no `C:\…`), and one shared `css/style.css` linked from every page	open it with Live Server	☐
12	Every link resolves: no 404 when you click	click every link on every page	☐

Submit: zip your folder (or push it to GitHub if you already use it) and write one line: *"The hardest item was __ because __."*

D2-HW3 · Self-Check & Interview Prep: Semantic HTML, Accessibility, Media → CSS Basics

File: homework/D2-HW3-Self-Check-and-Interview-Prep.md

TSI Nepal · AI-Powered Full Stack Developer Program · Week 2, Day 2 · Layer: Frontend · about 20 minutes

Mode: Theory + coding, because interviewers ask semantic and CSS questions viva-style ("What is the box model?", "id vs class?") and then hand you a small "style this card" or "why is this CSS not working?" task.

How to use this sheet

1. **Part 1** checks your memory. Answer without the notes, then mark ✔ / ~ / ✗.
2. **Part 2**: say the theory answers **out loud** (20–40 seconds each). Write the code answers, then **check them in the browser** with F12 → Styles and Computed.
3. Your instructor shares the answer key after you try.

How you'll be asked about this

- **Written / MCQ test:** "Which property adds space *inside* the border?", "Which selector has the highest specificity?", "Which tag is for the main content?"
- **Viva round:** "What is semantic HTML?", "What is the box model?", "Difference between `class` and `id`?", "Inline, block and inline-block?", "Three ways to add CSS?"
- **Machine test / take-home:** "Style this profile card", "Make this nav horizontal", "Fix this page so Lighthouse accessibility is above 90".
- **What they are really checking:** that you can *debug* CSS (find why a rule is ignored), not only write it. Saying "I inspected it and the rule was crossed out because…" wins interviews.

Topic map

Sub-topic	Why interviewers care	Questions
Semantic HTML5	SEO, accessibility, readable code	S1, S2, T1, T2
Accessibility and meta	Many companies now test it	S3, S4, T3
Audio, video, iframe	Common on real sites	S5
CSS rule and 3 ways to add CSS	The basics of every stylesheet	S6, T4, C4
Selectors and specificity	"Why is my style not applied?"	S7, T5, T6, C1
Colours, units, fonts	Valid values, `rem`	S8, C3
Box model	The #1 CSS interview question	S9, T7, C2
Display	Nav bars, hiding things	S10, T8, C5
Debugging CSS / Bootstrap	Junior speed	T9

Part 1: Self-check (10 questions)

S1. Name the semantic tag for: the top of the page, the menu of links, the main content, a blog post that makes sense on its own, a side box, the bottom of the page.

S2. Why is `<div class="btn">Subscribe</div>` a bad button? Give one thing you can test with your keyboard.

S3. Write a good `alt` for a photo of the TSI building, and the right `alt` for a decorative dhaka pattern strip.

S4. Name **three** things in the `<head>` that help a phone, a search engine or a screen reader.

S5. An `<audio>` tag shows nothing on the page. What attribute is missing? And what does a video's `poster` do?

S6. Inline style, a `<style>` block, and `<link rel="stylesheet">`. Which one is the **school uniform**, and why is it the best for a 4-page site?

S7. Write a selector for each: every `<p>`; everything with class `veg`; the element with id `todays-special`; only links inside `nav`; an element that has **both** classes `dish` and `spicy`.

S8. What is the difference between `px` and `rem`? Why do we give `font-family` a fallback?

S9. Draw (or describe) the box model from the inside out. Use the parcel words.

S10. What is the difference between `display: none` and `visibility: hidden`?

At the bottom of this file, write: `My ✗ topics:` _____

Part 2: Interview questions (14 questions)

Theory (say it out loud)

T1. (Warm-up) What is semantic HTML? Why does it matter?

T2. (Warm-up) When should you use `<section>` and when `<article>`? Can a page have two `<main>` elements?

T3. (Core) What does Lighthouse's Accessibility report check? Name four things that lower the score.

T4. (Core) What are the three ways to add CSS to a page? Which one wins if all three set the colour of the same `<h1>`?

T5. (Core) What is the difference between a `class` and an `id`? When would you use each?

T6. (Core) What is specificity? Rank these from weakest to strongest: `#menu`, `p`, `.price`.

T7. (Core) Explain the box model. What does `box-sizing: border-box` change?

T8. (Core) What is the difference between `block`, `inline` and `inline-block`? Why does `margin-top` do nothing on a link?

T9. (Stretch) "My CSS is not working." Walk me through how you would debug it, step by step.

Code (write it, then check it in the browser)

C1. Predict the colour. What colour is "Rs. 150"? Why? Would the answer change if `.price` came last in the file?

```
<p id="special" class="price">Rs. 150</p>
```

```
#special { color: green; }
.price   { color: red; }
p        { color: blue; }
```

C2. Predict the width. How wide is the card on screen (from the left edge of its border to the right edge)? And with `box-sizing: border-box`?

```
.card {
  width: 300px;
  padding: 20px;
  border: 5px solid #14213D;
  margin: 10px;
}
```

C3. Find the bugs. Four mistakes. For each one, say what the browser does with it.

✗ *Broken on purpose: find what is wrong*

```
h1 { color: #gg0000; font-size: 1.5 rem }
.card { padding: 20; border: 2px orange; }
```

C4. Find the bugs. Sunita says: "None of my CSS works, and the card class is ignored too." Find three problems.

✗ *Broken on purpose: find what is wrong*

```
<head>
  <link rel="stylsheet" href="css/styles.css">
</head>
<body>
  <div class="Card">Sunita Gurung</div>
</body>
```

The file on disk is `css/style.css`, and it contains `.card { border: 2px solid #990011; }`.

C5. Write it (5–10 minutes). Turn this list into a horizontal nav bar: no bullets, links side by side, each link with `10px 16px` padding, navy `#14213D` background, white text, and a cherry `#990011` background on hover. Don't change the HTML.

```
<nav>
  <ul class="menu">
    <li><a href="index.html">Home</a></li>
    <li><a href="menu.html">Menu</a></li>
    <li><a href="order.html">Order</a></li>
    <li><a href="about.html">About</a></li>
  </ul>
</nav>
```

Talk about your project

- "In your portfolio (D2-HW2), why did you use one external stylesheet for every page?"
- "Show me one element in DevTools and explain its box: content, padding, border, margin."
- "What was your Lighthouse accessibility score? What did you change to raise it?"

Coming later: you may still be asked

- **Flexbox and grid** ("How do you centre a div?"): sessions 3 / 3.2. Today's answer: `margin: 0 auto` with a width.
- **Positioning** (`relative`, `absolute`, `fixed`) and **media queries** (responsive design): sessions 3 / 3.2.
- **Bootstrap grid** (`row`, `col-md-6`): session 3.3.

Remember the One Rule: you may only ship code you can explain, line by line.

My ✗ topics: _____

D2-HW4 · Patan Craft Shop replay (stretch)

File: homework/D2-HW4-patan-craft-shop-replay/README.md

Time: +30 min (at home, optional) · **Type:** 🏴‍☠️ Bug-hunt

Goal: A last round: 10 mistakes from every Day 2 topic on a craft shop in Patan.

Open the folder in VS Code, then open `index.html` with Live Server.

Task

1. Look at the page in a normal window and in a narrow window (or F12 → device toolbar, Ctrl+Shift+M in DevTools).
2. Press play on both videos.
3. Run Lighthouse (Accessibility + SEO + Best practices).
4. Mend each mistake in `index.html` and `css/style.css`.

The bugs

There are 10 bugs. Find at least 8.

Hints, one per bug type:

- **The label on the parcel:** two `<meta>` tags are missing from `<head>`.
- **Media:** where must `<source>` go? Is its `type` real?
- **Selectors:** capital letters.
- **Values:** how many values can `padding` take? How many numbers does `rgb()` need?
- **Sizes:** more than 100% is wider than the window. A fixed height can be too small.
- **Hiding:** one way hides and keeps the space; the other removes the box.

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Done looks like

A brown header band, no gap under it, an orange banner that fits the window (no sideways scroll), a story card whose text stays inside, and two videos that both play.

Scoreboard

```
I found __ / 10. The reveal taught me: ________
```

Stuck?

1. Sideways scroll? F12 → Elements → hover the boxes: which one is wider than the window?
2. A video shows 0:00 and will not play? Read its `<source>` line very carefully.
3. Read `notes/09-debugging-css.md`.

Fast?

You have done every Day 2 hunt. Help a classmate, or polish D2-HW2.